

Differentiation Rules	
Derivative of a Constant Function	$\frac{d}{dx}(c) = 0$
$\frac{d}{dx}(x) = 1$	
The Power Rule (General Version) : If n is any real number , then	$\frac{d}{dx}(x^n) = nx^{n-1} , n \in \mathbb{R}$
The Constant Multiple Rule : If c is a constant $f(x)$ and $g(x)$ are both differentiable , then	$\frac{d}{dx}[cf(x)] = c \frac{d}{dx}f(x)$
The Sum Rule : If $f(x)$ and $g(x)$ are both differentiable , then	$\frac{d}{dx}[f(x) + g(x)] = \frac{d}{dx}f(x) + \frac{d}{dx}g(x)$
The Difference Rule : If $f(x)$ and $g(x)$ are both differentiable , then	$\frac{d}{dx}[f(x) - g(x)] = \frac{d}{dx}f(x) - \frac{d}{dx}g(x)$
The Product Rule : If $f(x)$ and $g(x)$ are both differentiable , then	$\frac{d}{dx}[f(x)g(x)] = g(x)\frac{d}{dx}f(x) + f(x)\frac{d}{dx}g(x)$ In prime notation : $(fg)' = fg' + gf'$
The Quotient Rule : If $f(x)$ and $g(x)$ are both differentiable , then	$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)\frac{d}{dx}f(x) - f(x)\frac{d}{dx}g(x)}{[g(x)]^2}$ In prime notation : $\left(\frac{f}{g}\right)' = \frac{fg' - gf'}{g^2}$
Derivative of the Natural Exponential Function	$\frac{d}{dx}e^x = e^x$ $\frac{d}{dx}e^{f(x)} = e^x \cdot (f(x))'$
Derivative of the Exponential Function	$\frac{d}{dx}a^x = a^x \ln(a)$
Definition of the Number e	
e is the number such that $\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1$	